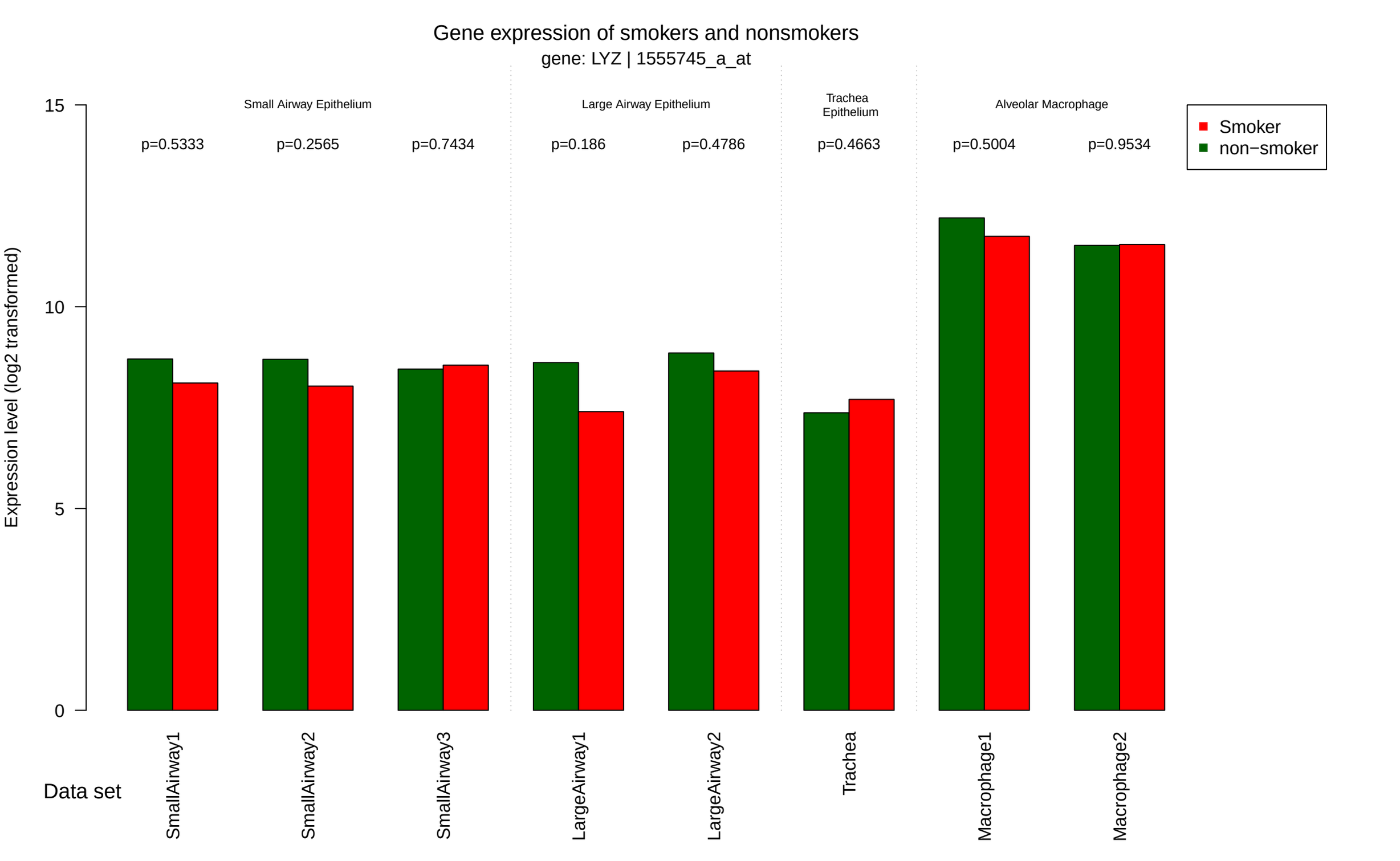
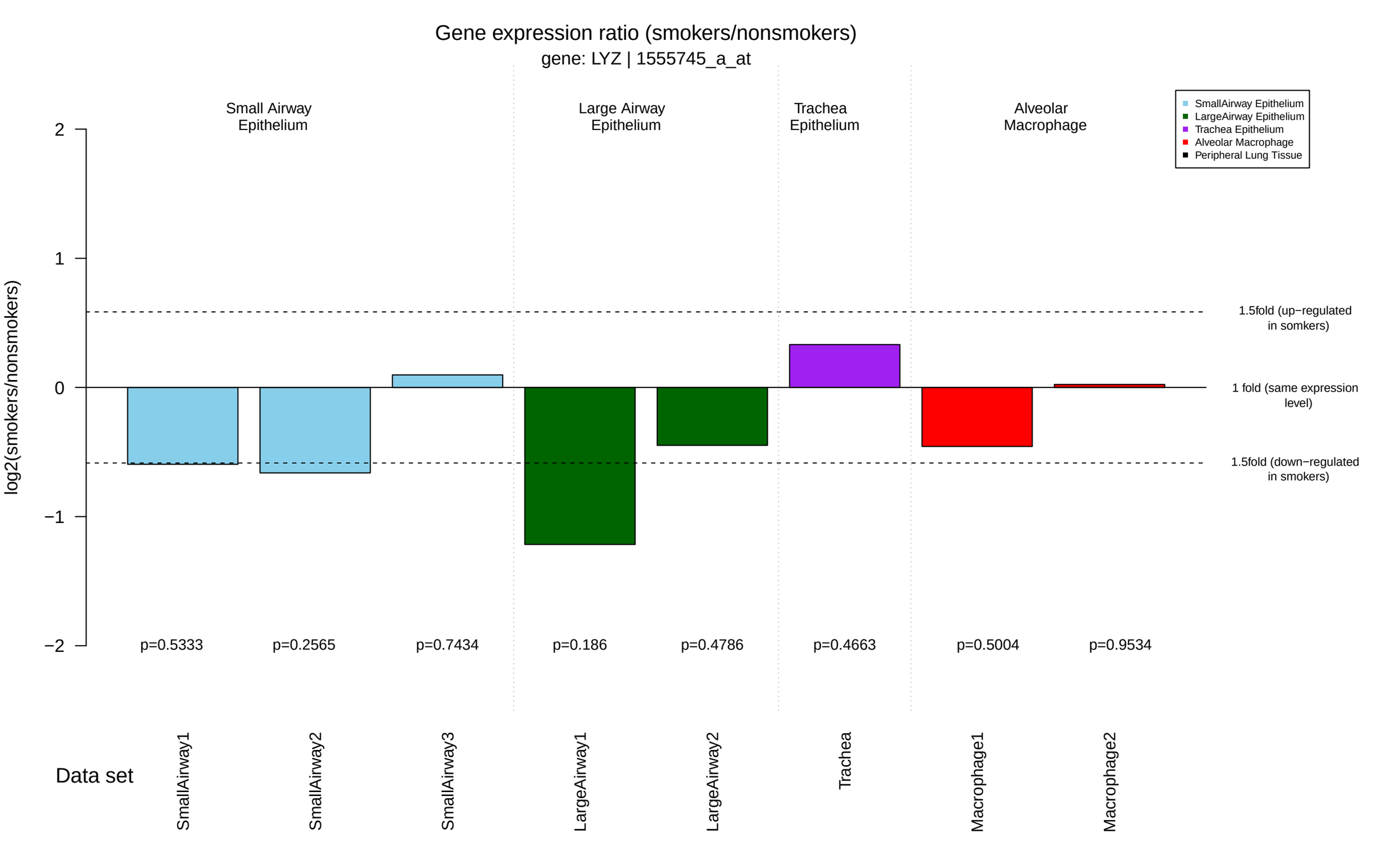
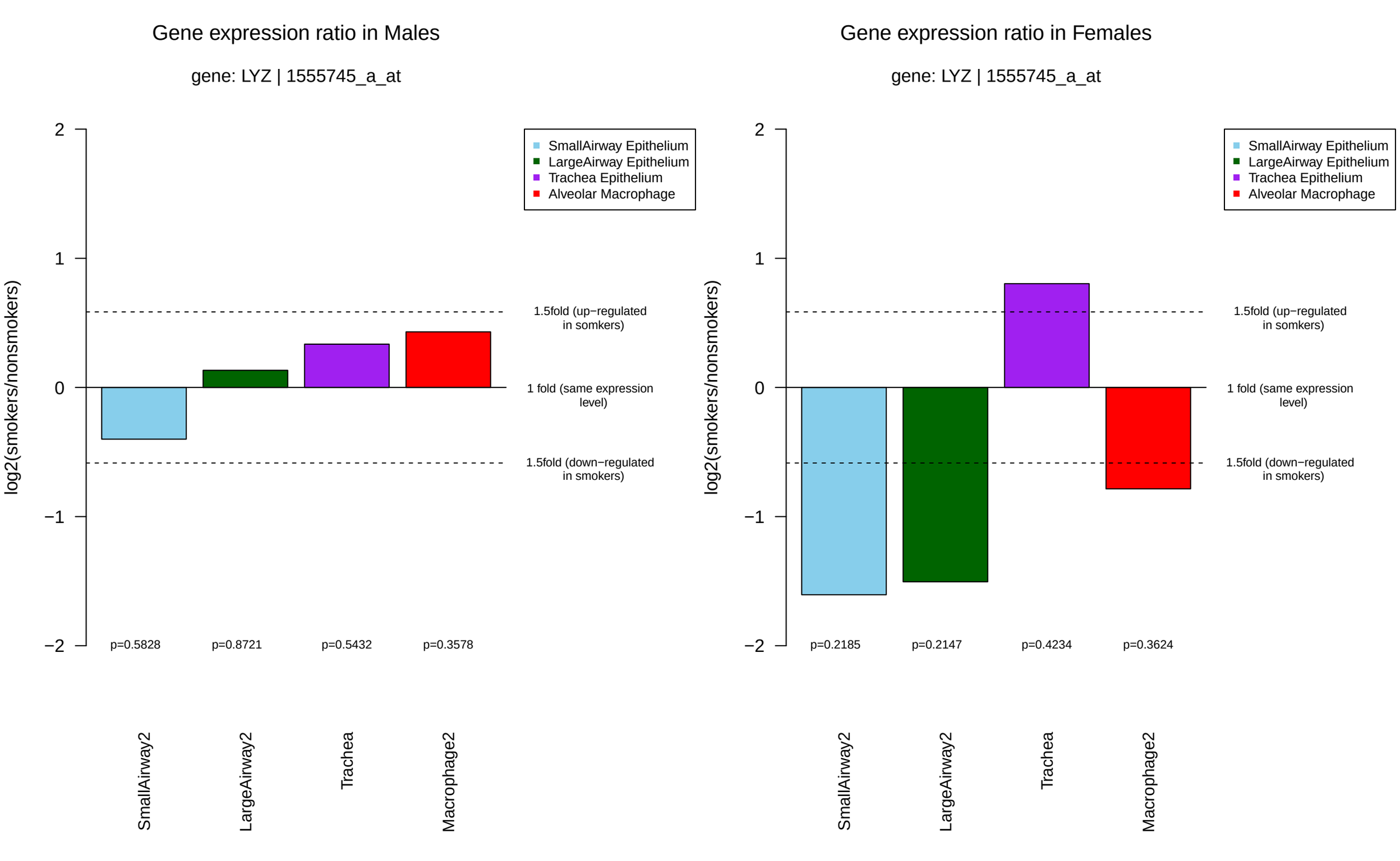
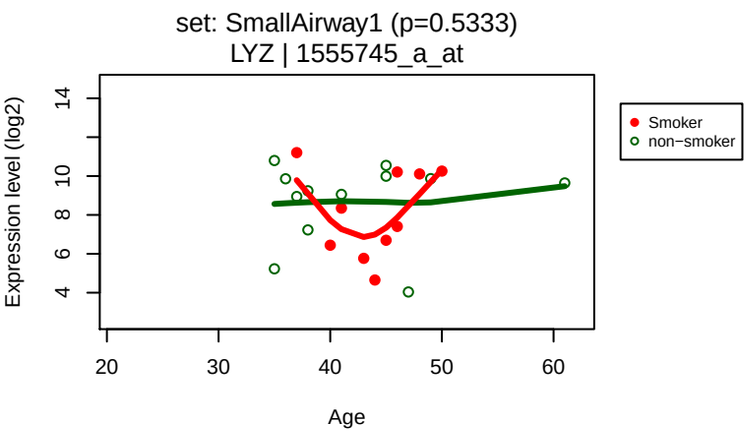
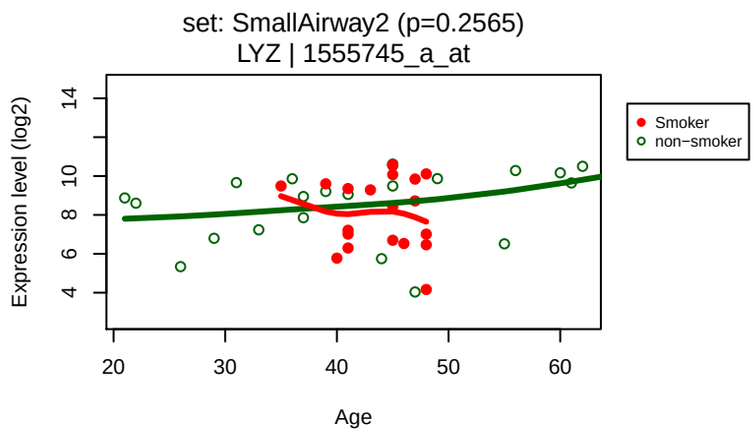
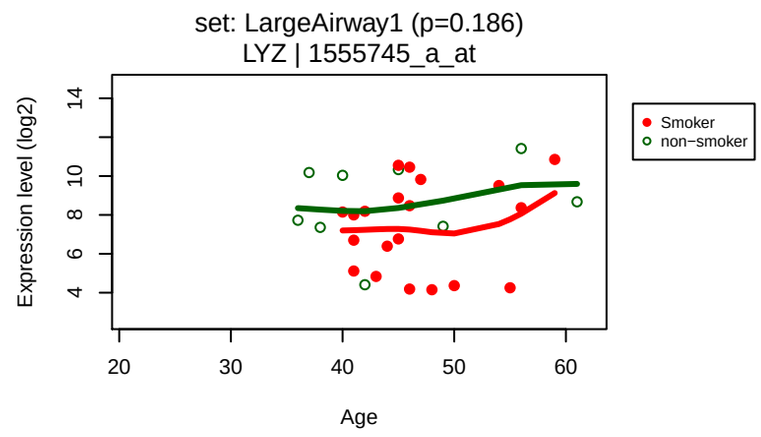

Gene expression of smokers and nonsmokers
gene: LYZ | 1555745_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5333
p=0.2565
p=0.7434
p=0.186
p=0.4786
p=0.4663
p=0.5004
p=0.9534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

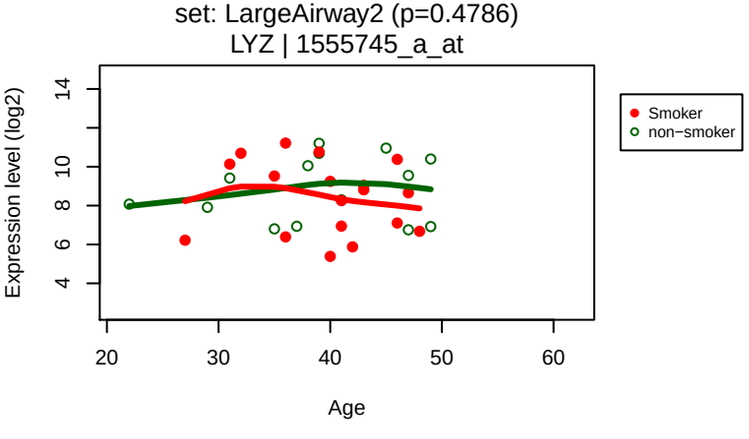

Gene expression ratio (smokers/nonsmokers)
gene: LYZ | 1555745_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5333
p=0.2565
p=0.7434
p=0.186
p=0.4786
p=0.4663
p=0.5004
p=0.9534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

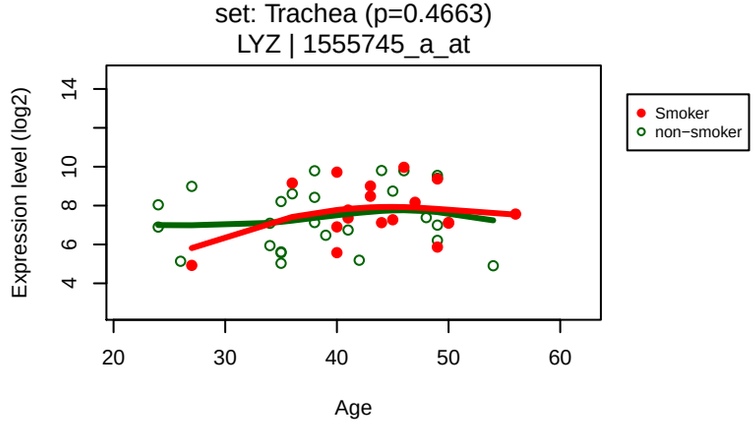

Gene expression ratio in Males
gene: LYZ | 1555745_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5828
p=0.8721
p=0.5432
p=0.3578
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LYZ | 1555745_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2185
p=0.2147
p=0.4234
p=0.3624
SmallAirway2
LargeAirway2
Trachea
Macrophage2

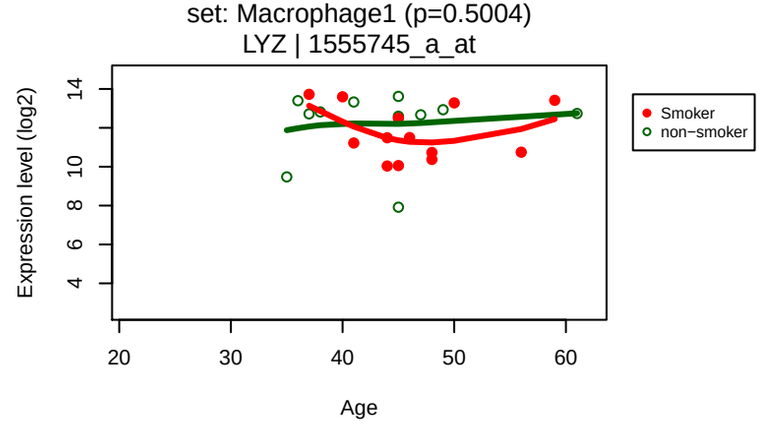

set: SmallAirway1 (p=0.5333)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker

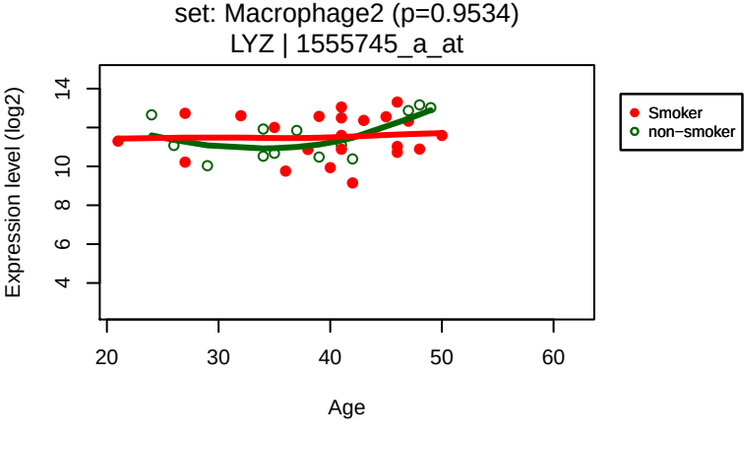

set: SmallAirway2 (p=0.2565)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker

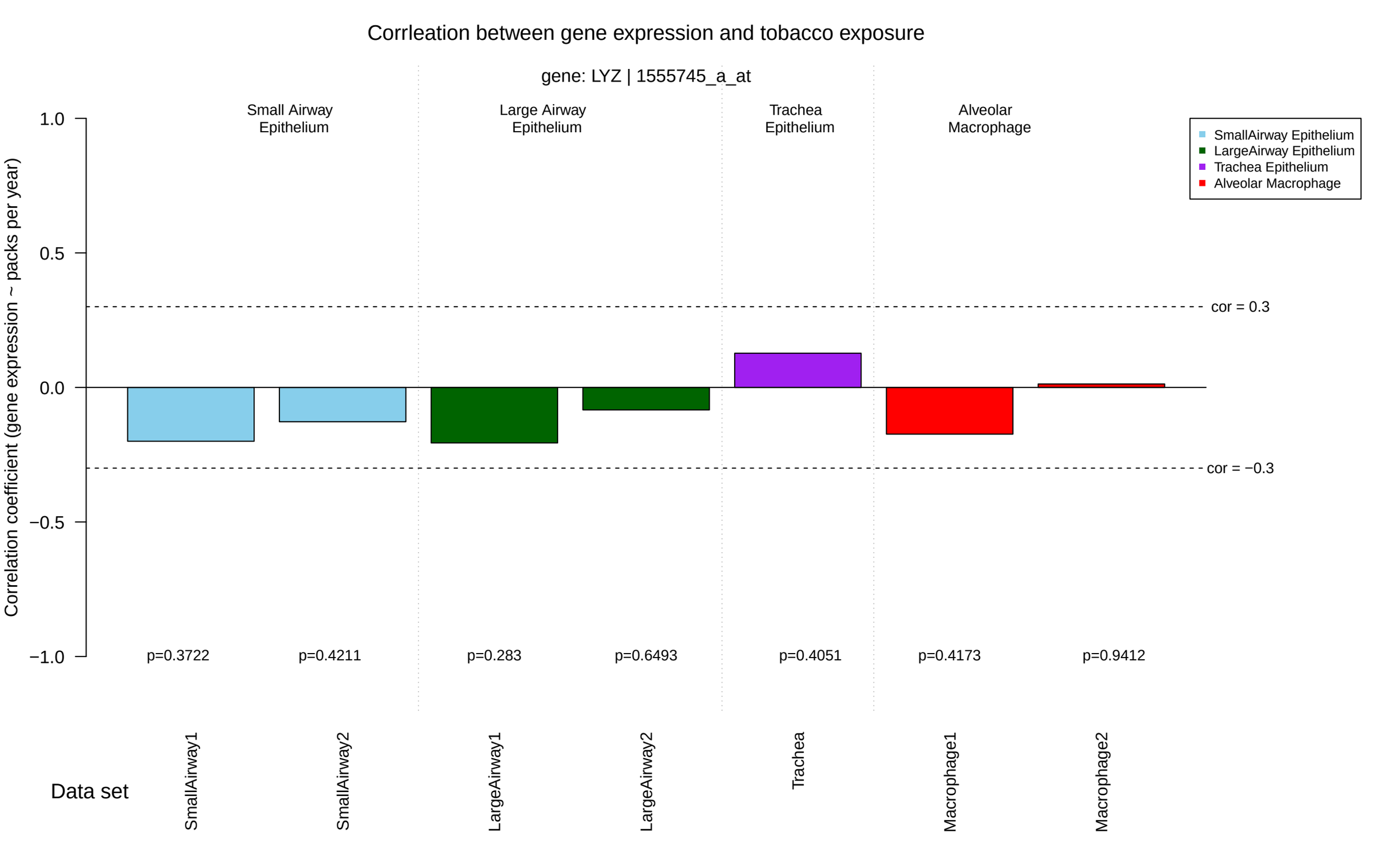

set: LargeAirway1 (p=0.186)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4786)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4663)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5004)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9534)
LYZ | 1555745_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LYZ | 1555745_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3722
p=0.4211
p=0.283
p=0.6493
p=0.4051
p=0.4173
p=0.9412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2